Final Report

Project Title: Stock Prediction using Python and ML

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Summary

Trading signals have long been used to identify potential opportunities in the market for short term trading. The project aims to use technical indicators on historical data of stocks to capture the gains provided by them. The trading signals used in the report are among the most widely used in the field of technical analysis. Through our Object Oriented Programming we aim to establish the historical returns provided by some of the common stocks with the help of technical indicators, such as SMA (Simple Moving Average), MACD (Moving Average Convergence/Divergence) and Bollinger Bands.

The returns and patterns deduced through this project can serve a practical purpose beyond retrospective analysis. As financial markets exhibit cyclical behaviour, by identifying which technical parameter historically generated favourable return, traders can establish trading rules when similar market conditions reappear.

In short, the project uses Python to build and test advanced trading strategies and deploy machine learning to forecast future prices.

Objectives

The objectives of this project are as follows:

1. Implementing a class to carry out technical analysis for a user input stock. The properties of the class will be as follows:
    - To get user's inputs on the stock for which they want to do the analysis on. The program will also want the user to give the start and end date.
    - To fetch price data from yahoo finance and use the closing prices to get trading signals using strategies such as SMA (Simple Moving Average), MACD (Moving Average Convergence/Divergence), Bollinger Bands. The program also calculates the return from each strategy.
    - The program is intended to identify potential opportunities in the market in terms of returns with trading indicators.
2. To feature engineer a wide range of predictive features from the raw data.
3. To design a flexible machine learning model class to forecast short-term stock price movements or direction.
4. To evaluate the performance of each model using the appropriate metrics, and identify which algorithm best serves to provide the most accurate and reliable forecasts for the given dataset.

Methodology/Code Execution

1. **Importing relevant libraries in our code:**
   Some of the libraries we used used in our projects are as follows:
    - `yfinance` - to get real time data from yahoo finance
    - `pandas` - to use dataframes for get trading signals and calculating returns
    - `numpy` - for calculating returns and handling mathematical computations

- `Matplotlib.pyplot, plotly.express, plotly.subplots` - plotting charts for technical analysis

For machine learning, we have used the following libraries:

- `seaborn`
- `unittest`
- `sklearn.model_selection import train_test_split`
- `sklearn.linear_model import LinearRegression, Ridge, Lasso`
- `sklearn.ensemble import RandomForestRegressor`
- `sklearn.svm import SVR`
- `sklearn.metrics import mean_squared_error, r2_score, mean_absolute_error`
- `from datetime import datetime, timedelta`

2. **Designing Relevant Classes for Streamlining**

The first step is to utilize object-oriented programming to its fullest to streamline the project. The following is a sample of the classes that we are in the process of designing to handle their respective aspects.

- ➔ `StockDataHandler()`: This class will handle all the data fetching and cleaning. Fetching data from the `yfinance` package in Python is essential to gather the data needed for later steps. The class has methods to download the data based on a list of stock tickers, to handle missing values, check data types, and return a clean DataFrame for a specific stock.
- ➔ `StockAnalyzer()`: This class is to perform financial analyses as described in our proposal. Some methods that are in this class are to calculate the daily returns, the volatility, risk metrics, correlation, and plotting the returns to visualize all of this information.
- ➔ `FeatureEngine()`: This class is to engineer technical indicators and features for the machine learning model. Some of the methods in this class are creating lagged features, adding the daily returns, adding the moving averages, adding trading volume changes, adding calendar features, etc. This also returns a final DataFrame ready to be adapted into the ML models.
- ➔ `MLModel()`: This class will handle the machine learning models. Some of the algorithms to be used are: linear regression, support vector machine, K-nearest neighbor, and random forest. The methods in this class will handle the logic for the algorithm, preprocess the DataFrame, train, predict, and evaluate how well the models perform.

Overall, by structuring the whole project around a few key classes, we will organize the project code.

3. **Class - StockDataHandler()**

As mentioned above, this class handles data fetching and cleaning from the `yfinance` package. It then stores specified stock data for use in other classes in the program. The

class can be checked through the subsequent `TestStockDataHandler()` class to ensure that it is working as intended.

- **`dl_data`**
  - This function downloads historical data for a specified stock via `yfinance`. It takes arguments for time period and interval (with a default of 1 day), then sorts the requested data by date.
- **`clean_data`**
  - This function cleans the downloaded data from `dl_data()`. It takes an argument for the type of fill the user would like to use to handle missing data, converts columns to numbers, and removes NaNs.
- **`get_clean_data`**
  - This function returns a copy of the cleaned data for the specified stock, and checks whether the requested data exists (that is, if it has been handled by the `clean_data()` function above). It will raise an error if the data is not found.
- **`get_data`**
  - This function acts as a wrapper for the `get_clean_data()` function. Its purpose is to call the copy of the cleaned, verified data for analysis.
- **`check_data_quality`**
  - This function checks for missing values in the specified data, and prints a list of them. `missing = df.isnull().sum().sum()` adds up all of the NaNs found in each column, to count how many NaNs total exist in the data.

4. **Class - StockAnalysis()**
   - **`__init__`**
     - The method serves as the constructor for our Stock Analysis Class. The parameters used are the ticker of the stock, start and end date. These parameters are stored as instance attributes to make them accessible throughout the class.
   - **`fetch_basic_statistics`**
     - The function retrieves fundamental company information from Yahoo Finance and displays them at the end of the report. The statistics reveal current market information of the company.
   - **`parse_mmddyyyy`**
     - The function standardizes user input date strings into pandas Timestamp objects with an explicit MM/DD/YYYY format specification so that there is a consistent input format throughout the class.
   - **`simple_analysis`**
     - The function handles the data acquisition and handling function within the class. One of the most common technical challenges of getting data from Yahoo Finance is that the API sometimes returns a MultiIndex DataFrame which makes the calculation of the technical returns and signals throw an error. The `isinstance()` function is used to handle any variation in

data structures that might occur due to different `yfinance` output formats. The function `.xs()` is used to flatten the data into a simple single column Data Frame with the closing prices. The simplified dataframe contains only the closing price serving as a convenient structure for plotting functions later in the class. The pandas dataframe structure requires a proper DatetimeIndex to enable specific operations like rolling window calculations used for trading analysis.

- **`find_return`**
  - The class has a program which returns the return of the stock between the start and the end date. The return will simply be used to compare with the return provided by the technical analysis and benchmark it as “buy and hold return”. The program used specific functions such as the `get_indexer()` method with backward fill to automatically map requested dates to the nearest actual trading day ensuring that weekend or holiday date requests automatically roll forward to the next trading day.
- **`interactive_plot`**
  - The function contains interactive line chart visualization of the stock’s closing prices using `plotly` Express library. The user can hover over the chart to get the closing price details of each day. The `isinstance()` function is again used here to make sure that the data structure is a simple single-column DataFrame
- **`candlestick_plot`**
  - One of the most used trading charts is a candlestick chart. It is different from the interactive chart that we just drew. It takes other information such as open, close, high and low price data of the stock, which helps us the trader in understanding the direction of the market.
- **`trading_support_resistance`**
  - Support and Resistance indicators are trend and momentum indicators. A trade is made on the assumption that a high price tends traders to sell the stocks and converse. Hence we calculate a support level as a recent price “floor” and a resistance level as a recent price “ceiling.”The level of support and resistance is calculated by taking the maximum and minimum price and adding a 20% margin. The high and low prices are dynamic in this case. The function operates on a rolling window basis (bin width). The function makes use of arrays to capture if there needs to be a trade (long or short). The function.loc is used to ensure proper index alignment. The support and resistance indicators are then plotted on a graph of the price movement. For this project the support and resistance indicators just provide the overall volatility of the stock and if we can trade in it, hence we do not calculate return through this method.
- **`SMA_indicator`**
  - Simple moving average is a simple technical indicator, which takes the average of the prices within a rolling window. It is easy to observe that the

20 day (longer SMA) SMA evens out the micro volatility in the market. In this trading strategy, we will buy the stock if the SMA-5 (shorter SMA) value is greater than SMA -20 price.

$$SMA = \frac{\sum_{i=1}^{N} Pi}{N}$$

The method implements a dual moving average crossover with a default 5 and 20 day window. We make a local copy of self.data as we do not want to overwrite or change the original dataframe. We create and use a completely different dataframe for all the trading strategies where the dataframe is isolated within the function's scope and do not alter the self.data dataframe. The function uses columns to identify the trading signals and calculate the return. The function uses a log of daily prices to calculate the return. At last the function calculates total and annual return of the stock. We also plot the chart to visually represent our trading strategy.

- **`macd_trading`**
  - The MACD indicator is similar to the SMAindicator in the sense that it catches the difference between a fast moving and slow moving trading indicator. However, in MACD we apply a smoothing exponential moving average.

$$MACD = EMA_{fast} - EMA_{fast}$$
$$MACD_{signal} = EMA_{MACD}$$

- **`bbands_trading`**
  - Bollinger Bands strategy also builds up on moving averages, but it also incorporates price volatility. Bollinger Bands plot a middle moving average with an upper and lower band set a chosen number of standard deviations away, so the band width expands in volatile periods and contracts when volatility is low.Typical settings use a 20-period simple moving average with bands at ±2 standard deviations, but both the lookback and multiplier can be tuned to the asset and timeframe. This is one of the most used strategies as it provides price adaptive price boundaries that adjust to the prices. The code uses a similar column method used in the above two trading strategies.
- **`run_all_simple_analysis`**
  - The function executes complete analysis methods in a logical sequence, so that the user is able to call just one function.
- **`summary`**
  - The function provides a single view of the trading strategy performance metrics.

5. **Class - FeatureEngine()**

The Feature Engine class serves a vital purpose for machine learning. Its primary purpose is to convert the raw financial data into a structured dataset with various predictive signals that can be used to train and test with a machine learning model. The class operates using Pandas DataFrame and applies mathematical applications to transform the data. The class accepts the target stock data and a dictionary of macroeconomic variables which are also dataframes. A local copy is created so the original data is immutable.

A more detailed description of the factors can be found above in the `Stock Analysis` class, however, these are the features that we will be using:

- Technical Indicators:
  - Simple Moving Average (SMA_10, SMA_50)
  - Relative Strength Index (RSI)
  - MACD (Moving Average Convergence Divergence)
  - ROC (Rate of Change)
- Volatility Indicators
  - Bollinger Bands
- Calendar Effects:
  - Is_Monday
  - Is_Friday
  - Month
- Macroeconomic Factors:
  - S&P 500
  - U.S. Oil
  - Volatility Index (VIX)

The above features each have a dedicated method to handle fetching or calculating the specific factors. Additionally, we have a target generation method using `prepare_target`. Since this specific machine learning task is supervised learning, we created a target column by shifting the closing prices backwards by a day. This means that “Today’s Data” (features) will align with “Tomorrow’s Price (target), which teaches the models to forecast the future based on the present.

6. **Class - MLModel()**

The ML class acts as a wrapper for the SciKit-Learn library’s estimators. It handles the preparation for testing and training data and evaluation. The following methods indicate how the class operates as follows:

- <u>Data Splitting and Scaling</u>: The method `split_data` splits the data into two pieces – the training data and testing data. Typically in machine learning applications, we reserve the last 20% of the data for testing purposes. A crucial note is to set “`shuffle = FALSE`”, to account for the fact that financial data is strictly sequential, so we cannot train 2023 data to predict 2021 prices. We also scale the data using `StandardScaler()`, as the dataset contains features with varying scales. For example, the price could be approximately $400, however the returns could be 0.05. To account for this discrepancy, especially for machine

learning models such as Ridge and SVM (which bias towards large numbers), we scale the features to have a mean of 0 and standard deviation of 1. Another crucial note is to implement this only on the training dataset to avoid data leakage, and artificially inflating the model's results.

- Model Training: The method `train_model` allows the user to switch between the algorithms by passing a simple string argument. The models we use are LinearRegression, Ridge, Lasso, RandomForest, and SVM. For Ridge and Lasso, we have alpha parameters, which introduces the regularization parameters which control the strength of the penalty assigned to any single feature, which will help to prevent overfitting of the model.
- Evaluation: The method `evaluate`, calculates popular metrics used to understand the performance of the machine learning algorithms. These metrics include Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the R^2 score. Each of these metrics help us understand different ways the model performed:
    - RMSE: This is the standard deviation of the prediction errors. It will heavily penalize outliers.
    - MAE: This is the average absolute difference between the predicted and actual prices.
    - R^2 Score: This represents the proportion of variance in the target variable that is predicted by the features. A score of 1.0 is perfect while 0.0 is equivalent to guessing the mean.

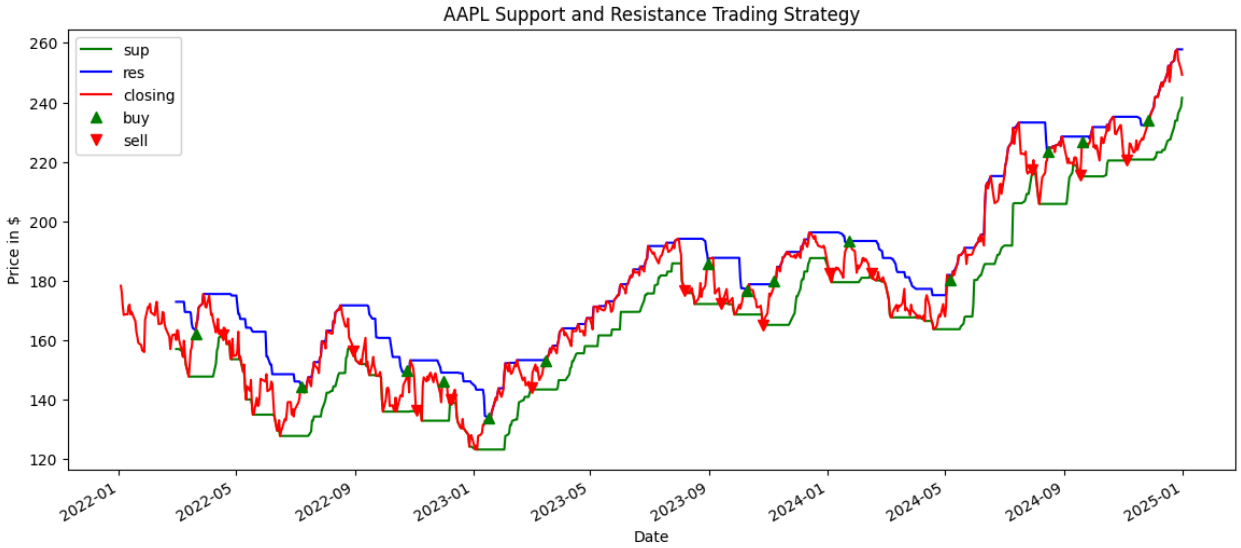
  The method ends with returning the prediction as well as `y_test`, which is the actual true values of the target variable from the testing portion of the data (i.e. an answer key to better evaluate the model with).

7. Results & Conclusions

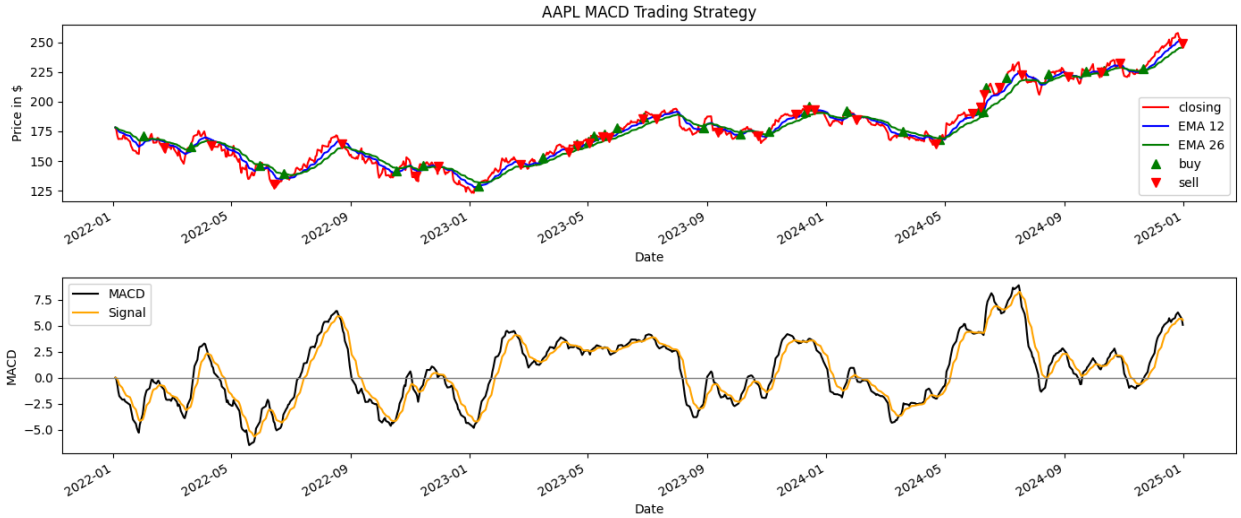
The output for the `StockAnalysis` class gives the following results:

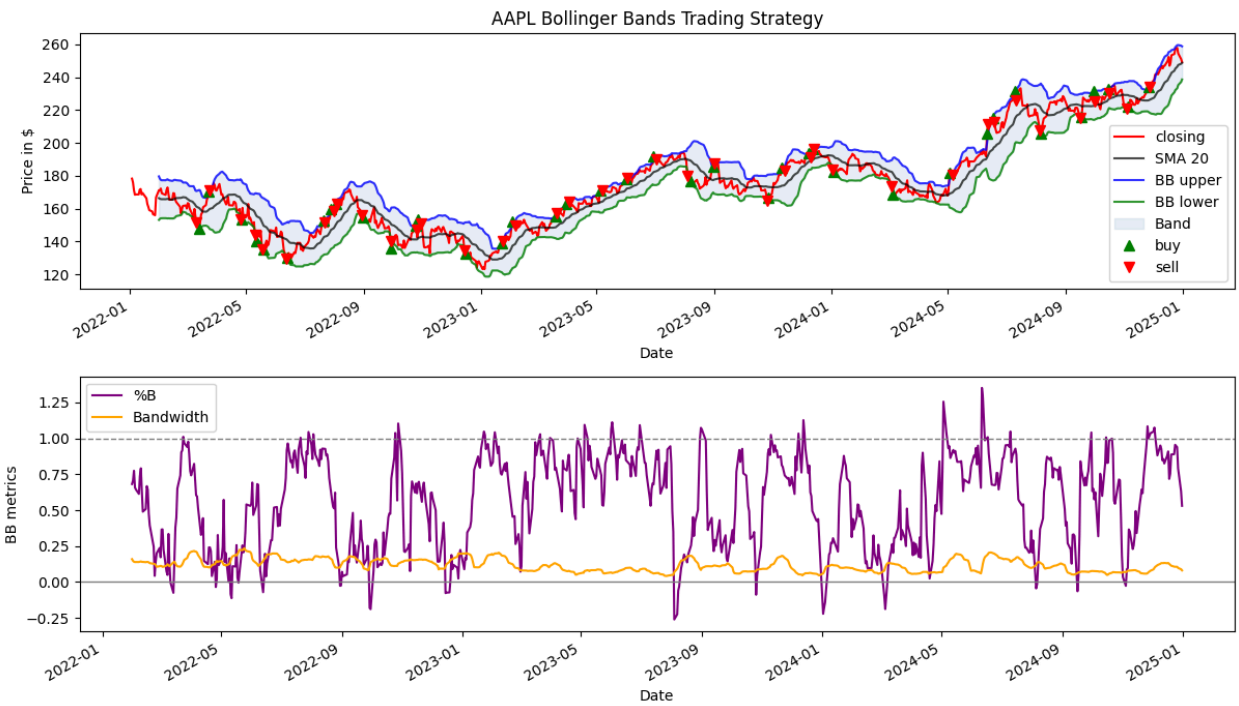
```
===== TRADING SUMMARY=====
Ticker: AAPL
Period: 2022-01-01 00:00:00 to 2025-01-01 00:00:00
Buy and Hold return (%): 39.84
SMA strategy total return (%): 46.64
SMA strategy annual return (%): 13.69
MACD strategy total return (%): 48.43
MACD strategy annual return (%): 14.15
Bollinger Bands strategy total return (%): 4.42
Bollinger Bands strategy annual return (%): 1.46
====================
```

The trading indicators of Apple Stock from various trading strategies are:


AAPL Support and Resistance Trading Strategy
sup
res
closing
buy
sell
Price in $
Date

AAPL SMA strategy
Close
SMA-5
SMA-20
Buy
Sell


AAPL MACD Trading Strategy
closing
EMA 12
EMA 26
buy
sell
Price in $
Date
MACD
Signal
MACD
Date



We evaluated the five aforementioned models (LinearRegression, Ridge, Lasso, RandomForest, and SVM) on the test dataset using APPL stock prices. The results highlight a distinct, yet not surprising, performance gap between the linear and non-linear models for this financial data task.

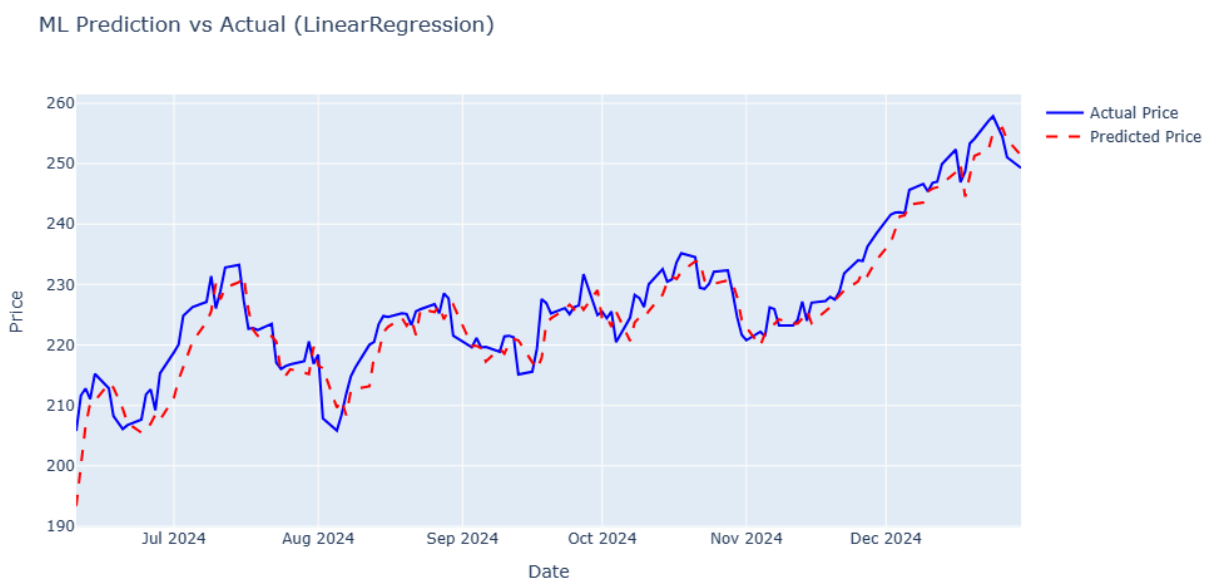
Model	RMSE	MAE	R^2 Score
Linear Regression	3.6339	2.8545	0.8982
Ridge Regression	3.6378	2.8095	0.8979
Lasso Regression	3.9692	3.0517	0.8785
Random Forest	39.8883	38.4950	-11.2715
SVM	64.0434	63.0169	-30.6341

Above, the high performing linear models are highlighted in green, and the failed models are highlighted in red.

- **<u>Strong Performance Models</u> - Linear Regression and Ridge Regression**
    - Both models achieved an R^2 score of almost 90%, which speaks to the accuracy of these two linear models. This high performance can be attributed to the fact that linear models determine a slope and intercept which allows them to project a high upwards trend (such as in Apple stock), indefinitely.

- Even when the price reached high prices, the linear regression models successfully predicted the higher prices. Ridge Regression also provided the lowest MAE of 2.8095. This indicated that, on average, the model's prediction was off by approximately $2.81 per share. This is due to the impact that regularization has on the noise from highly volatile features, which led to slightly more stable predictions as compared to Linear Regression.

- **<u>Weak Performance Models</u> - Random Forest and SVM**
  - Unsurprisingly, these models were unfit for a financial time series dataset. As Random Forest works by averaging the values of similar past data points, it caps the predictions to the highest training value it receives, making it unfit for financial time series predictions where stock prices could always be on the rise. This is demonstrated by the huge errors with an RMSE of 39.9 and an R^2 of -11.2715.
  - SVM performed even worse than Random Forest, given that SVM relies on vector distance, and the new test data points were far outside the vector space of the training data, resulting in the model losing its predictive power.



The graph above highlights the best performing model – Linear Regression.

As observed in the graph, the predicted price closely follows the trend of the actual price, thereby reinforcing all the aforementioned observations.

In conclusion, this project allowed us to demonstrate that for predicting stock prices using machine learning models, linearity, such as with Linear Regression and Ridge Regression, serves better and outperforms complex non-linear algorithms such as Random Forest and SVM.

The OOP design of this project allowed us to quickly gather, analyze, and evaluate these models.

Challenges

- ➔ The data from Yahoo Finance throwing Multiframe Index errors.
  - ◆ This was solved by using the `isinstance()` function.
- ➔ Getting datetime objects to format correctly.
  - ◆ This was solved by using the `datetime` package.
- ➔ Understanding the performances of the ML models.
  - ◆ This was solved by researching how to interpret the RMSE, MAE, and R^2.

Future Work Challenges

Future analysis of stock data using our model will require more sophisticated and efficient code, in order to perform the requisite calculations, and this is beyond the current scope of the project. As more modules and features are added, the code will also need to be quality checked to ensure no functionality has been lost, no new bugs have arisen, and consistency is maintained.

To support more rigorous analyses, our code should eventually become scalable to handle multiple tickers at once in order to compare them. Robust data handling, such as errors while fetching, API throttling, and download failures will also be necessary, potentially via caching.

Furthermore, in regard to machine learning applications, given that the dataset is a financial time sensitive dataset, time series algorithms such as ARIMA or GARCH should also be explored and compared alongside the linear regression models, in order to compare and contrast the results each algorithm would provide.

Working Code

```
import yfinance as yf
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import unittest
import plotly.express as px
import plotly.graph_objects as go
from plotly.subplots import make_subplots
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, Ridge, Lasso
from sklearn.ensemble import RandomForestRegressor
from sklearn.svm import SVR
from sklearn.metrics import mean_squared_error, r2_score,
mean_absolute_error
```

```
from datetime import datetime, timedelta

class StockDataHandler:
    def __init__(self):
        self.raw_data = {}
        self.cleaned_data = {}
          self.tickers = [] # list to store tickers for the StockAnalysis
class to use

    def dl_data(self, tickers, start=None, end=None, interval="1d"):
        self.tickers = tickers
        for t in tickers:
              df = yf.download(t, start=start, end=end, interval=interval,
progress=False, auto_adjust=True)
            if df.empty:
                continue
            df.index = pd.to_datetime(df.index)
            df = df.sort_index()
            self.raw_data[t] = df
    def clean_data(self, fill_method="ffill"):
        for t, df in self.raw_data.items():
            d = df.copy()
            if isinstance(d.columns, pd.MultiIndex):
                d.columns = d.columns.droplevel(1)
            d.columns = d.columns.astype(str)
            if fill_method == "ffill":
                d = d.ffill().bfill()
            elif fill_method == "bfill":
                d = d.bfill().ffill()
            elif fill_method == "drop":
                d = d.dropna()
            for col in d.columns:
                d[col] = pd.to_numeric(d[col], errors="coerce")
            d = d.dropna()
            self.cleaned_data[t] = d

    def get_clean_data(self, ticker):
        if ticker not in self.cleaned_data:
              raise ValueError(f"Data for {ticker} not found. Run dl_data
and clean_data first.")
```

```
        return self.cleaned_data[ticker].copy()

    def get_data(self, ticker):
        return self.get_clean_data(ticker)

    def check_data_quality(self):
        for ticker, df in self.cleaned_data.items():
            missing = df.isnull().sum().sum()
             print(f"Ticker: {ticker} | Missing Values: {missing} | Shape:
{df.shape}")

class TestStockDataHandler(unittest.TestCase):

    def setUp(self):
         self.ticker = "AAPL" #change this to the stock ticker you want to
test
        self.handler = StockDataHandler()

        end = datetime.today()
        start = end - timedelta(days=60)

        self.handler.dl_data(
            [self.ticker],
            start=start.strftime("%Y-%m-%d"),
            end=end.strftime("%Y-%m-%d"),
            interval="1d"
        )
        self.handler.clean_data()

    def test_raw_data_stored(self):
        self.assertIn(self.ticker, self.handler.raw_data)
        df_raw = self.handler.raw_data[self.ticker]
        self.assertGreater(len(df_raw), 0)

    def test_clean_data_stored(self):
        self.assertIn(self.ticker, self.handler.cleaned_data)
        df_clean = self.handler.cleaned_data[self.ticker]
        self.assertGreater(len(df_clean), 0)

    def test_get_clean_data_returns_copy(self):
```

```
        df_clean = self.handler.get_clean_data(self.ticker)
        self.assertGreater(len(df_clean), 0)

        old_shape = self.handler.cleaned_data[self.ticker].shape
        df_clean["DummyColumn"] = 1
        new_shape_internal = self.handler.cleaned_data[self.ticker].shape
        self.assertEqual(old_shape, new_shape_internal)

    def test_get_clean_data(self):
        df1 = self.handler.get_clean_data(self.ticker)
        df2 = self.handler.get_data(self.ticker)
        self.assertTrue(df1.equals(df2))

    def test_check_data_quality_runs(self):
        self.handler.check_data_quality()


if __name__ == "__main__":
       unittest.main(argv=['first-arg-is-ignored'], exit=False)

##Defining a Stock Analysis Class

class Stock_Analysis():

   ##Creating the constructor function and initialize the attributes
    def __init__(self, ticker, start_date, end_date):
        self.ticker = str(ticker)
        self.start_date = start_date
        self.end_date = end_date

        self.data = None
        self.data_df_close = None
        self.lowest_prices = None
        self.return_value = None

    ##Defining a function to fetch basic statistics from Yahoo Finance
    def fetch_basic_statitiscs(self):
        ticker_stats = yf.Ticker(self.ticker)
```

```
        info = ticker_stats.info  # creating a dictionary to store the
fundamentals

        stats = {
            "longName": info.get("longName"),
            "sector": info.get("sector"),
            "industry": info.get("industry"),
            "marketCap": info.get("marketCap"),
            "beta": info.get("beta"),
            "trailingPE": info.get("trailingPE"),
            "forwardPE": info.get("forwardPE"),
            "dividendYield": info.get("dividendYield"),
            "trailingEps": info.get("trailingEps"),
            "currentPrice": info.get("currentPrice"),
            "52WeekHigh": info.get("fiftyTwoWeekHigh"),
            "52WeekLow": info.get("fiftyTwoWeekLow")}

        self.basic_stats = stats
        return stats

    ##Defining a function so that the inputs are in the correct format as
MM/DD/YYYY
    def parse_mmddyyyy(self, s: str) -> pd.Timestamp:
        # Interpreting '1/12/2024' as MM/DD/YYYY (Jan 12, 2024)
        return pd.to_datetime(s, format="%m/%d/%Y")

     ##Defining a function to download the data from yahoo finance in a
dataframe for easy use and handle MultiIndex columns
    def simple_analysis(self):
        ticker = self.ticker
        start = self.parse_mmddyyyy(self.start_date)
        end   = self.parse_mmddyyyy(self.end_date)
        data = yf.download(ticker, start, end)

         # In case yfinance returned MultiIndex columns, select the ticker
and flatten
        if isinstance(data.columns, pd.MultiIndex):
        # to take the subcolumns for this ticker, e.g. ('Close','AAPL') ->
'Close'
```

```
              data = data.xs(ticker, axis=1, level=1) ##creating a simple
dataframe for the specific ticker
                      data.columns  =  data.columns.astype(str)    #  now:
'Open','High','Low','Close','Adj Close','Volume'

        ##Storing the data and making columns for analysis
        self.data = data
        self.data.index = pd.to_datetime(self.data.index)

        # these now work because 'Close' is a normal column
        self.data["closing"] = self.data["Close"]
        self.data["daily_change"] = self.data["closing"].diff()

        # Closing prices only with column name as ticker
        self.data_df_close = self.data[["Close"]].rename(columns={"Close":
self.ticker})


      ##Defining an function to find the return between the two entered
dates and ensuring the dates exist in the index
    def find_return(self, start_date, end_date):
        data = self.data_df_close

        start_date = self.parse_mmddyyyy(start_date)
        end_date = self.parse_mmddyyyy(end_date)
        ticker = self.ticker

              #  Ensuring  that  the  dates  exist  in  index  in  case  of
weekends/holidays
             start_date  =  data.index[data.index.get_indexer([start_date],
method="bfill")[0]]
              end_date     =  data.index[data.index.get_indexer([end_date],
method="bfill")[0]]

        start_price = data.loc[start_date, ticker]
        end_price   = data.loc[end_date,   ticker]
        returns = ((end_price - start_price) / start_price) * 100
        returns = round(returns, 2)

        self.return_value = {
```

```
            "ticker": ticker,
            "start_date": start_date,
            "end_date": end_date,
            "return_pct": returns,
        }
        return returns

     ##Defining a function to create an interactive plot of the closing
price of the chosen stock price
    def interactive_plot(self, title="Interactive prices"):
        df = self.data_df_close
           ##Checking with isinstance if the dataframe is multi index or
series and converting it to a single column dataframe if it is so
              if isinstance(df, pd.DataFrame) and isinstance(df.columns,
pd.MultiIndex):
            df = df["Close"].to_frame(name=self.ticker)
        elif isinstance(df, pd.Series):
            df = df.to_frame(name=self.ticker)

        print("-------Interactive Plot-------")
         print("Hover around the graph to see the closing prices for each
date!!!")

        fig = px.line(df, x=df.index, y=df.columns[0], title=title)
        fig.show()

    def candlestick_plot(self):
        # working on a local copy of self.data
        data = self.data.copy()

        # Handle MultiIndex columns from yfinance
        if isinstance(data.columns, pd.MultiIndex):
            data = data.xs(self.ticker, axis=1, level=1)

        # Standardize column names and ensure datetime index
        data.columns = data.columns.str.strip().str.title()
        data.index = pd.to_datetime(data.index).tz_localize(None)

        # Now these columns exist
        data = data.dropna(subset=["Open", "High", "Low", "Close"])
```

```
        print("-------Candlestick Plot-------")
          print("Interactive candlestick chart for opening, high, low and
closing price analysis")
        # Plotly candlestick
        fig = go.Figure(
            [
                go.Candlestick(
                    x=data.index,
                    open=data["Open"],
                    high=data["High"],
                    low=data["Low"],
                    close=data["Close"],
                )
            ]
        )
        fig.update_layout(
            template="plotly_white",
            xaxis_rangeslider_visible=False,
            title=f"{self.ticker} Candlestick Chart",
        )
        fig.show()
        print("-------Interactive Plot for Volume-------")
        print("Interactive plot for volume for",self.ticker)
        fig2 = make_subplots(specs=[[{"secondary_y": True}]])
        fig2.add_trace(
            go.Scatter(x=data.index, y=data["Close"], name="Price"),
            secondary_y=False,
        )
        fig2.add_trace(
            go.Bar(x=data.index, y=data["Volume"], name="Volume"),
            secondary_y=True,
        )
        fig2.update_layout(title=f"{self.ticker} Price and Volume")
        fig2.show()

    ##Defining a function for support and resistance trading strategy
    def trading_support_resistance(self, bin_width=20):
        # work on a copy of self.data for safety
        data = self.data.copy()
        n = len(data)
```

```
        ##Creating arrays to store the calculated values for the strategy
        support_tolerance   = np.full(n, np.nan)
        resistance_tolerance = np.full(n, np.nan)
        support_count   = np.zeros(n, dtype=int)
        resistance_count = np.zeros(n, dtype=int)
        support_arr     = np.full(n, np.nan)
        resistance_arr  = np.full(n, np.nan)
        signal          = np.zeros(n, dtype=int)

        in_support = 0
        in_resist  = 0

        price = data['closing'].to_numpy()

        ##Calculating support and resistance levels along with the trading
signals
        for x in range((bin_width - 1) + bin_width, n):
            low = x - bin_width + 1
            high = x + 1
            window = price[low:high]
            sup = window.min()
            res = window.max()
            rng = res - sup

            support_arr[x] = sup
            resistance_arr[x] = res
            support_tolerance[x] = sup + 0.2 * rng
            resistance_tolerance[x] = res - 0.2 * rng

            if resistance_tolerance[x] <= price[x] <= resistance_arr[x]:
                in_resist += 1
                resistance_count[x] = in_resist
                in_support = 0
            elif support_tolerance[x] >= price[x] >= support_arr[x]:
                in_support += 1
                support_count[x] = in_support
                in_resist = 0
            else:
                in_support = 0
                in_resist = 0
```

```
            if in_resist >= 2:
                signal[x] = 1
            elif in_support >= 2:
                signal[x] = 0
            else:
                signal[x] = signal[x - 1]

        # assign back once, index-aligned
        data.loc[:, 'sup_tolerance'] = support_tolerance
        data.loc[:, 'res_tolerance'] = resistance_tolerance
        data.loc[:, 'sup_count']     = support_count
        data.loc[:, 'res_count']     = resistance_count
        data.loc[:, 'sup']           = support_arr
        data.loc[:, 'res']           = resistance_arr
        data.loc[:, 'signal']        = signal
        data.loc[:, 'positions']     = data['signal'].diff()

                 print("-------Support  and  Resistance  Trading  Strategy
Plot-------")
        print("Support and resistance strategy for", self.ticker)

        fig = plt.figure(figsize=(14, 6))
        ax1 = fig.add_subplot(111, ylabel='Price in $')
            ax1.set_title(f'{self.ticker} Support and Resistance Trading
Strategy')
        data['sup'].plot(ax=ax1, color='green')
        data['res'].plot(ax=ax1, color='blue')
        data['closing'].plot(ax=ax1, color='red')
        ax1.plot(
            data.loc[data.positions == 1.0].index,
            data.closing[data.positions == 1.0],
            '^', markersize=7, color='green', label='buy'
        )
        ax1.plot(
            data.loc[data.positions == -1.0].index,
            data.closing[data.positions == -1.0],
            'v', markersize=7, color='red', label='sell'
        )
        plt.legend()
```

```
        plt.show()

        # store results for summary
        self.sr_last_signal = data['signal'].iloc[-1]
        self.sr_last_position = data['positions'].iloc[-1]
        return data

    ## simple moving average tradig strategy
    def SMA_indictor(self, window= 5, window_1= 20):
        ##making a copy of the data for safety
        df = self.data[['Close']].copy()
        df.rename(columns={'Close': 'Close'}, inplace=True)

        # defining fast and slow SMA
                     df['SMA_fast']   =   df['Close'].rolling(window=window,
min_periods=1).mean()
                    df['SMA_slow']   =   df['Close'].rolling(window=window_1,
min_periods=1).mean()

        # shift by 1 day to avoid look-ahead
        df['SMA_fast_shift'] = df['SMA_fast'].shift(1)
        df['SMA_slow_shift'] = df['SMA_slow'].shift(1)

        # trading signal: long = 1, short = -1, 0 = flat
                     df['signal_SMA']   =   np.where(df['SMA_fast_shift']   >
df['SMA_slow_shift'], 1, 0)
                     df['signal_SMA']   =   np.where(df['SMA_fast_shift']   <
df['SMA_slow_shift'],
                                  -1, df['signal_SMA'])

        # log returns and strategy returns
        df['log_return'] = np.log(df['Close']).diff()
        df['log_return_trend'] = df['signal_SMA'] * df['log_return']

        # entry / exit points
        df['action'] = df['signal_SMA'].diff()
        cum_factor = np.exp(df['log_return_trend'].sum())
        total_return = cum_factor - 1 # as decimal
        total_return_pct = round(total_return * 100, 2)
```

```
        print("-------Simple Moving Average Trading Strategy Plot-------")
        print("SMA strategy for", self.ticker)
           print("Total SMA strategy return for the entered dates (%):",
total_return_pct)
        self.sma_total_return_pct = total_return_pct
        n_days = df['log_return_trend'].count()
        ann_factor = 252
        annual_return = (1 + total_return) ** (ann_factor / n_days) - 1 if
n_days > 0 else np.nan
        self.sma_annual_return_pct = round(annual_return * 100, 2)
          print("Annual SMA strategy return for the entered dates (%):",
self.sma_annual_return_pct)

        # plot
        plt.figure(figsize=(14, 6))
        plt.grid(True, alpha=0.3)
        plt.plot(df.index, df['Close'], label='Close')
        plt.plot(df.index, df['SMA_fast'], label=f'SMA-{window}')
        plt.plot(df.index, df['SMA_slow'], label=f'SMA-{window_1}')
        plt.plot(df.loc[df['action'] == 2].index,
            df['SMA_fast'][df['action'] == 2],
            '^', color='g', markersize=12, label='Buy')
        plt.plot(df.loc[df['action'] == -2].index,
            df['SMA_slow'][df['action'] == -2],
            'v', color='r', markersize=12, label='Sell')
        plt.legend(loc=1)
        plt.title(f'{self.ticker} SMA strategy')
        plt.show()

        # store last signal for summary
        self.sma_last_signal = int(df['signal_SMA'].iloc[-1])
        self.sma_total_return_pct = round(total_return * 100, 2)
        self.sma_annual_return_pct = round(annual_return * 100, 2)
        return df

    def macd_trading(self, fast_span=12, slow_span=26, sig_span=9):
       ##creating a copy of the data
        data_v1 = self.data[['closing']].copy()

        # defiing EMAs for trading strategy
```

```
            data_v1['EMA_fast']  =  data_v1['closing'].ewm(span=fast_span,
adjust=False).mean()
            data_v1['EMA_slow']  =  data_v1['closing'].ewm(span=slow_span,
adjust=False).mean()

        # getting a MACD line, Signal line
        data_v1['MACD'] = data_v1['EMA_fast'] - data_v1['EMA_slow']
             data_v1['MACD_signal']  =  data_v1['MACD'].ewm(span=sig_span,
adjust=False).mean()

        # trading rule: MACD crosses Signal by using yesterday's values to
avoid look-ahead
        data_v1['MACD_shift'] = data_v1['MACD'].shift(1)
        data_v1['MACD_signal_shift'] = data_v1['MACD_signal'].shift(1)
        data_v1['signal'] = np.where(
        data_v1['MACD_shift'] > data_v1['MACD_signal_shift'], 1,
        np.where(data_v1['MACD_shift'] < data_v1['MACD_signal_shift'], -1,
0))
        data_v1['positions'] = data_v1['signal'].diff()  # +2 buy, -2 sell

        data_v1['log_return'] = np.log(data_v1['closing']).diff()
        data_v1['signal_shift'] = data_v1['signal'].shift(1).fillna(0)   #
use yesterday's signal
            data_v1['macd_log_return_trend']  =  data_v1['signal_shift']  *
data_v1['log_return']

        print("-------MACD Trading Strategy Plot-------")
        print("MACD strategy for", self.ticker)
        # total return of MACD strategy
        macd_cum_factor = np.exp(data_v1['macd_log_return_trend'].sum())
        macd_total_return = macd_cum_factor - 1
        macd_total_return_pct = round(macd_total_return * 100, 2)
        n_days = data_v1['macd_log_return_trend'].count()
        ann_factor = 252
          macd_annual_return  =  ((1  +  macd_total_return)  **  (ann_factor  /
n_days)  - 1 if n_days > 0 else np.nan)
        macd_annual_return_pct = round(macd_annual_return * 100, 2)
```

```
          print("Total MACD strategy return for the entered dates (%):",
macd_total_return_pct)
          print("Annual MACD strategy return for the entered dates(%):",
macd_annual_return_pct)

        self.macd_total_return_pct = macd_total_return_pct
        self.macd_annual_return_pct = macd_annual_return_pct

        # plot
        fig = plt.figure(figsize=(14, 6))
        ax1 = fig.add_subplot(211, ylabel='Price in $')
        ax1.set_title(f'{self.ticker} MACD Trading Strategy')
        data_v1['closing'].plot(ax=ax1, color='red', label='closing')
              data_v1['EMA_fast'].plot(ax=ax1,  color='blue',  label=f'EMA
{fast_span}')
             data_v1['EMA_slow'].plot(ax=ax1,  color='green',  label=f'EMA
{slow_span}')
        ax1.plot(data_v1.loc[data_v1.positions == 2].index,
        data_v1.closing[data_v1.positions == 2],
        '^', markersize=7, color='green', label='buy')
        ax1.plot(
        data_v1.loc[data_v1.positions == -2].index,
        data_v1.closing[data_v1.positions == -2],
        'v', markersize=7, color='red', label='sell')
        ax1.legend(loc='best')

        ax2 = fig.add_subplot(212, ylabel='MACD')
        data_v1['MACD'].plot(ax=ax2, color='black', label='MACD')
                     data_v1['MACD_signal'].plot(ax=ax2,   color='orange',
label='Signal')
        ax2.axhline(0.0, color='gray', lw=1)
        ax2.legend(loc='best')
        plt.tight_layout()
        plt.show()

        self.macd_last_signal = int(data_v1['signal'].iloc[-1])
        return data_v1

    ##Defining a function for Bollinger Bands trading strategy
    def bbands_trading(self, window=20, k=2):
```

```
        # making a copy of the data
        data_v1 = self.data[['closing']].copy()

        # middle band (SMA), rolling std, upper/lower bands
                  data_v1['BB_mid']  =  data_v1['closing'].rolling(window,
min_periods=window).mean()
                  data_v1['BB_std']  =  data_v1['closing'].rolling(window,
min_periods=window).std()
        data_v1['BB_up']  = data_v1['BB_mid'] + k * data_v1['BB_std']
        data_v1['BB_dn']  = data_v1['BB_mid'] - k * data_v1['BB_std']

        close_shift = data_v1['closing'].shift(1)
        up_shift    = data_v1['BB_up'].shift(1)
        dn_shift    = data_v1['BB_dn'].shift(1)

          cross_up    = (close_shift <= up_shift) & (data_v1['closing'] >
data_v1['BB_up'])
          cross_down = (close_shift >= dn_shift) & (data_v1['closing'] <
data_v1['BB_dn'])
         ##Generating trading signals based on Bollinger Bands crossovers,
long = 1, short = -1, flat = 0
        data_v1['signal'] = np.where(cross_up, 1, np.where(cross_down, -1,
0))
        data_v1['positions'] = data_v1['signal'].diff()  # +1 buy, -1 sell

        # ----- returns for BB strategy -----
        data_v1['log_return'] = np.log(data_v1['closing']).diff()
        data_v1['signal_shift'] = data_v1['signal'].shift(1).fillna(0)   #
look-ahead avoidance
              data_v1['bb_log_return_trend']  =  data_v1['signal_shift']  *
data_v1['log_return']

        # total and annualized return
        bb_cum_factor = np.exp(data_v1['bb_log_return_trend'].sum())
        bb_total_return = bb_cum_factor - 1
        bb_total_return_pct = round(bb_total_return * 100, 2)

        n_days = data_v1['bb_log_return_trend'].count()
        ann_factor = 252
```

```
        bb_annual_return = ((1 + bb_total_return) ** (ann_factor / n_days)
- 1) if n_days > 0 else np.nan
        bb_annual_return_pct = round(bb_annual_return * 100, 2)

        print("-------Bollinger Bands Trading Strategy Plot-------")
        print("Bollinger Bands strategy for", self.ticker)
           print("Total Bollinger Bands strategy return for entered dates
(%):", bb_total_return_pct)
              print("Annual Bollinger Bands strategy return for entered
dates(%):", bb_annual_return_pct)

        self.bb_total_return_pct = bb_total_return_pct
        self.bb_annual_return_pct = bb_annual_return_pct

        # ----- plot for Bollinger Band Strategy -----
        fig = plt.figure(figsize=(12, 7))
        ax1 = fig.add_subplot(211, ylabel='Price in $')
        ax1.set_title(f'{self.ticker} Bollinger Bands Trading Strategy')
        data_v1['closing'].plot(ax=ax1, color='red', label='closing')
                data_v1['BB_mid'].plot(ax=ax1,  color='black',  alpha=0.7,
label=f'SMA {window}')
          data_v1['BB_up'].plot(ax=ax1, color='blue', alpha=0.8, label='BB
upper')
         data_v1['BB_dn'].plot(ax=ax1, color='green', alpha=0.8, label='BB
lower')

                       ax1.fill_between(data_v1.index,   data_v1['BB_dn'],
data_v1['BB_up'],
                     color='steelblue', alpha=0.12, label='Band')

        ax1.plot(data_v1.loc[data_v1.positions == 1].index,
             data_v1.closing[data_v1.positions == 1], '^', markersize=7,
             color='green', label='buy')
        ax1.plot(data_v1.loc[data_v1.positions == -1].index,
             data_v1.closing[data_v1.positions == -1], 'v', markersize=7,
             color='red', label='sell')
        ax1.legend(loc='best')

        ax2 = fig.add_subplot(212, ylabel='BB metrics')
```

```
            data_v1['BB_pctB'] = (data_v1['closing'] - data_v1['BB_dn']) /
(data_v1['BB_up'] - data_v1['BB_dn'])
            data_v1['BB_bw']    = (data_v1['BB_up'] - data_v1['BB_dn']) /
data_v1['BB_mid']
        data_v1['BB_pctB'].plot(ax=ax2, color='purple', label='%B')
        data_v1['BB_bw'].plot(ax=ax2, color='orange', label='Bandwidth')
        ax2.axhline(0.0, color='gray', lw=1)
        ax2.axhline(1.0, color='gray', lw=1, ls='--')
        ax2.legend(loc='best')
        plt.tight_layout()
        plt.show()

        self.bb_df = data_v1
        return data_v1


    def run_all_simple_analysis(self):
        print("-----BASIC STOCK ANALYSIS-----")
        print("Running all analysis for", self.ticker)

        self.simple_analysis()
        self.find_return(self.start_date, self.end_date)
        self.interactive_plot('Interactive prices for')
        self.candlestick_plot()
        self.trading_support_resistance()
        self.SMA_indictor()
        self.macd_trading()
        self.bbands_trading()
        self.fetch_basic_statitiscs()
        self.summary()


    def summary(self):

        print("\n===== SUMMARY =====")
        if hasattr(self, "basic_stats"):
            s = self.basic_stats
            print("\nBasic statistics from Yahoo Finance:")
            if s.get("longName"):
                print("  Name:", s["longName"])
```

```
            if s.get("sector"):
                print("  Sector:", s["sector"])
            if s.get("industry"):
                print("  Industry:", s["industry"])
            if s.get("marketCap") is not None:
                print("  Market cap:", s["marketCap"])
            if s.get("trailingPE") is not None:
                print("  Trailing P/E:", s["trailingPE"])
            if s.get("beta") is not None:
                print("  Beta:", s["beta"])

        print("\n===== TRADING SUMMARY=====")

        print("Ticker:", self.ticker)
        print("Period:", self.start_date, "to", self.end_date)
        if self.return_value is not None:
                                print("Buy   and   Hold   return   (%):",
self.return_value["return_pct"])
        if hasattr(self, "sma_total_return_pct"):
                                print("SMA   strategy   total   return   (%):",
self.sma_total_return_pct)
        if hasattr(self, "sma_annual_return_pct"):
                                print("SMA   strategy   annual   return   (%):",
self.sma_annual_return_pct)
        if hasattr(self, "macd_total_return_pct"):
                                print("MACD   strategy   total   return   (%):",
self.macd_total_return_pct)
        if hasattr(self, "macd_annual_return_pct"):
                                print("MACD   strategy   annual   return   (%):",
self.macd_annual_return_pct)
        if hasattr(self, "bb_total_return_pct"):
                    print("Bollinger  Bands  strategy  total  return  (%):",
self.bb_total_return_pct)
        if hasattr(self, "bb_annual_return_pct"):
                    print("Bollinger  Bands  strategy  annual  return  (%):",
self.bb_annual_return_pct)

        print("===================\n")

class FeatureEngine:
```

```
  def __init__(self, df, macro_data=None):
        self.df = df.copy()
        # macro_data in a dict: {'SPY': df, 'Oil': df, ...}
        self.macro_data = macro_data

  def add_technical_indicators(self):
        # Adds SMA, RSI, MACD, ROC
        # SMA (10 and 50)
        self.df['SMA_10'] = self.df['Close'].rolling(window=10).mean()
        self.df['SMA_50'] = self.df['Close'].rolling(window=50).mean()

        # RSI (14) - Relative Strength Index
        delta = self.df['Close'].diff()
        gain = (delta.where(delta > 0, 0)).rolling(window=14).mean()
        loss = (-delta.where(delta < 0, 0)).rolling(window=14).mean()
        rs = gain / loss
        self.df['RSI'] = 100 - (100 / (1 + rs))

        # MACD - Moving Average Convergence Divergence
        ema12 = self.df['Close'].ewm(span=12, adjust=False).mean()
        ema26 = self.df['Close'].ewm(span=26, adjust=False).mean()
        self.df['MACD'] = ema12 - ema26
                    self.df['MACD_Signal']   =   self.df['MACD'].ewm(span=9,
adjust=False).mean()

        # ROC - Rate of Change over 10 days
        self.df['ROC'] = self.df['Close'].pct_change(periods=10) * 100

        return self

  def add_bollinger_bands(self, window=20, num_std=2):
        # Adds Bollinger Bands (Upper and Lower) to the dataframe
        rolling_mean = self.df['Close'].rolling(window=window).mean()
        rolling_std = self.df['Close'].rolling(window=window).std()

        self.df['BB_Upper'] = rolling_mean + (rolling_std * num_std)
        self.df['BB_Lower'] = rolling_mean - (rolling_std * num_std)
        return self

  def add_calendar_features(self):
```

```
        # Adds Monday/Friday flags and Month seasonality.
        # Is Monday
        self.df['Is_Monday'] = (self.df.index.dayofweek == 0).astype(int)

        # Is Friday
        self.df['Is_Friday'] = (self.df.index.dayofweek == 4).astype(int)

        # Month of Year
        self.df['Month'] = self.df.index.month

        # Start of Month - Days 1 to 5
        self.df['Is_Start_Month'] = (self.df.index.day <= 5).astype(int)

        # Middle of Month  - Days 14 to 16
                self.df['Is_Mid_Month']  =  ((self.df.index.day  >=  14)  &
(self.df.index.day <= 16)).astype(int)

  def add_macro_features(self):
        # Merges external macro factors like S&P500, VIX, Oil.
        if not self.macro_data:
            return self

        for name, macro_df in self.macro_data.items():

            macro_series = macro_df['Close'].rename(f'{name}_Close')

                    self.df  =  self.df.merge(macro_series,  left_index=True,
right_index=True, how='left')

            if name in ['SP500', 'Oil', 'Treasury', 'VIX']:
                                                self.df[f'{name}_Return']   =
self.df[f'{name}_Close'].pct_change()

        return self

  def prepare_target(self):
        # Creates Next Day's Close.
        self.df['Target'] = self.df['Close'].shift(-1)
        return self.df.dropna()
```

```
class MLModel:
    def __init__(self, data, feature_cols, target_col='Target'):
        self.data = data
        self.feature_cols = feature_cols
        self.target_col = target_col
        self.model = None
        self.scaler = StandardScaler()
            self.X_train, self.X_test, self.y_train, self.y_test = (None,
None, None, None)

    def split_data(self, test_size=0.2, shuffle=False):
        X = self.data[self.feature_cols]
        y = self.data[self.target_col]

                self.X_train, self.X_test, self.y_train, self.y_test =
train_test_split(
            X, y, test_size=test_size, shuffle=shuffle
        )
          print(f"Data Split: Train shape {self.X_train.shape}, Test shape
{self.X_test.shape}")

    def train_model(self, model_type='LinearRegression'):
        if model_type == 'LinearRegression':
            self.model = LinearRegression()
        elif model_type == 'RandomForest':
                    self.model = RandomForestRegressor(n_estimators=100,
max_depth=10, random_state=42)
        elif model_type == 'SVM':
            self.model = SVR(kernel='rbf', C=100, gamma=0.1, epsilon=0.1)
        elif model_type == 'Ridge':
            self.model = Ridge(alpha=1.0)
        elif model_type == 'Lasso':
            self.model = Lasso(alpha=0.1)
        else:
            raise ValueError("Unknown model type")

        self.model.fit(self.X_train, self.y_train)
        print(f"Model {model_type} trained.")

    def evaluate(self):
```

```
        predictions = self.model.predict(self.X_test)
        mse = mean_squared_error(self.y_test, predictions)
        rmse = np.sqrt(mse)
        r2 = r2_score(self.y_test, predictions)
        mae = mean_absolute_error(self.y_test, predictions)
        print(f"Evaluation Metrics ({self.model.__class__.__name__})")
        print(f"RMSE: {rmse:.4f}")
        print(f"MAE: {mae:.4f}")
        print(f"R2 Score: {r2:.4f}")

        return predictions, self.y_test

if __name__ == "__main__":

  # Define stock

  target_stock = "AAPL"
  start_date = pd.to_datetime("2022-01-01")
  end_date = pd.to_datetime("2025-01-01")

  macro_tickers = {
      'SP500': '^GSPC',
      'Treasury': '^TNX',
      'Oil': 'USO',
      'VIX': '^VIX'
      }

  all_tickers = [target_stock] + list(macro_tickers.values())

  # Fetch Data

  handler = StockDataHandler()
  handler.dl_data(all_tickers, start = start_date, end = end_date)
  handler.clean_data()

  stock_df = handler.get_data(target_stock)
    macro_data = {name : handler.get_data(ticker) for name, ticker in
macro_tickers.items()}

  # Stock Analysis
```

```
stock_analyzer = Stock_Analysis(target_stock, start_date, end_date)
stock_analyzer.run_all_simple_analysis()

# Feature Engineering

engine = FeatureEngine(stock_df, macro_data = macro_data)
engine.add_technical_indicators()
engine.add_bollinger_bands()
engine.add_calendar_features()
engine.add_macro_features()

ml_data = engine.prepare_target()

print(f"Final Data Shape: {ml_data.shape}")

# Machine Learning Implementation

 feature_cols = ['SMA_10', 'SMA_50','RSI', 'MACD', 'MACD_Signal', 'ROC',
'BB_Upper', 'BB_Lower', 'Is_Monday', 'Is_Friday', 'Month', 'SP500_Return',
'Treasury_Return', 'Oil_Return', 'VIX_Return']

ml_model = MLModel(ml_data, feature_cols)
ml_model.split_data()

models = ['LinearRegression', 'RandomForest', 'SVM', 'Ridge', 'Lasso']

best_rmse = float('inf')
best_preds = None
best_model_name = ""
actuals = None

for model_name in models:
    print(f"\n--- Testing {model_name} ---")
    ml_model.train_model(model_name)
    preds, y_test = ml_model.evaluate()
    current_rmse = np.sqrt(mean_squared_error(y_test, preds))
    if current_rmse < best_rmse:
        best_rmse = current_rmse
        best_preds = preds
```

```
          best_model_name = model_name
          actuals = y_test

  results_df = pd.DataFrame({
        'Date': actuals.index,
        'Actual': actuals.values,
        'Predicted': best_preds
    })

  fig = go.Figure()
    fig.add_trace(go.Scatter(x=results_df['Date'],  y=results_df['Actual'],
name='Actual Price', line=dict(color='blue')))
                              fig.add_trace(go.Scatter(x=results_df['Date'],
y=results_df['Predicted'],  name='Predicted Price',  line=dict(color='red',
dash='dash')))
   fig.update_layout(title=f'ML Prediction vs Actual ({best_model_name})',
xaxis_title='Date', yaxis_title='Price')
  fig.show()

  # Unit Testing
 suite = unittest.TestLoader().loadTestsFromTestCase(TestStockDataHandler)
 unittest.TextTestRunner(verbosity=2).run(suite)
```